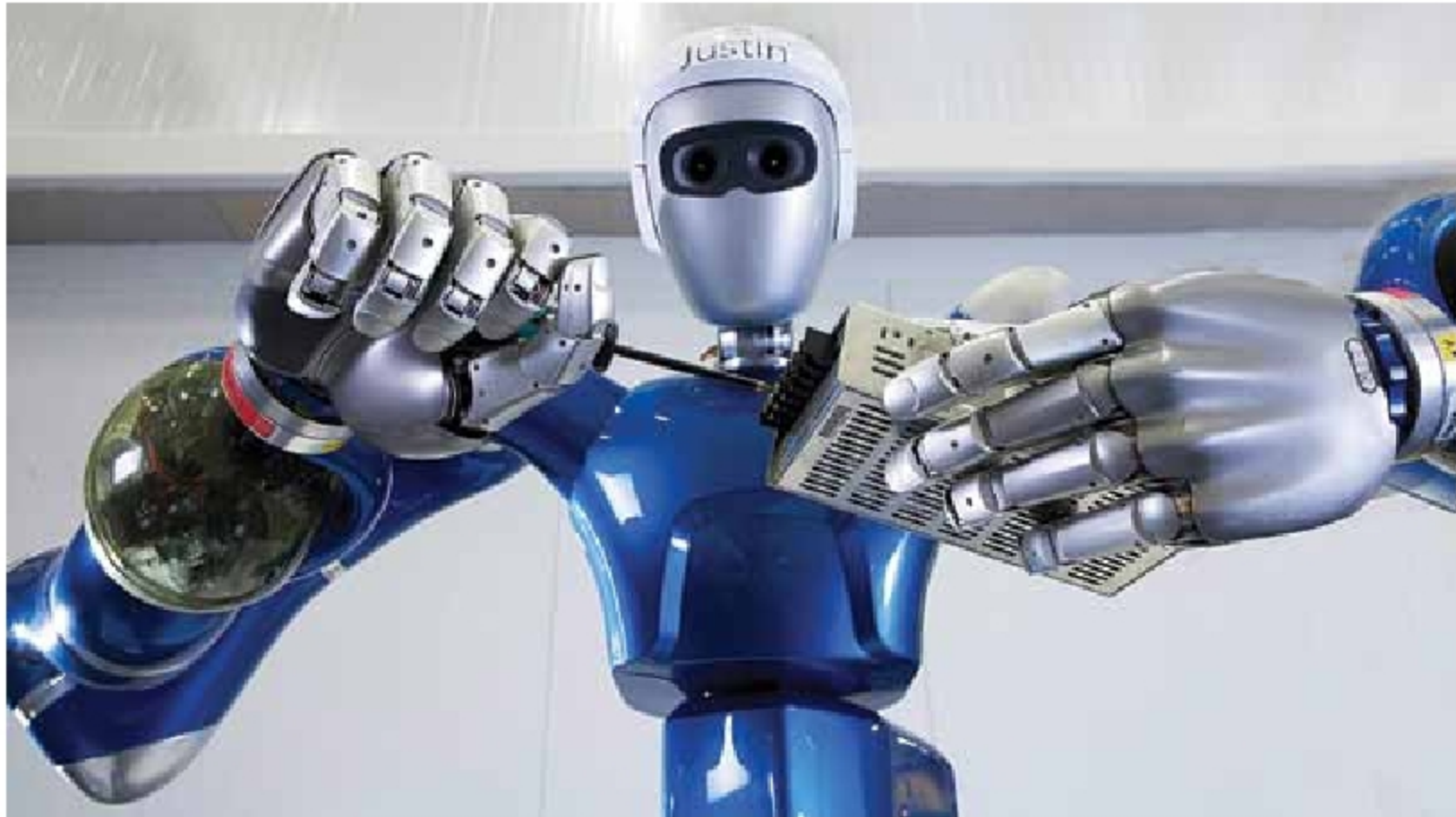


AI in E-Waste Management: Dig Gold and Burn the Shit!

The future of smart recycling is looking brighter. Robots, guided by cameras and artificial intelligence, are helping make recycling facilities run more efficiently.



Representation of AI-powered robot (Credit: spectrum.ieee.org)

You must have come across the joke “God created heaven and earth; the rest was made in China.” But, what about the garbage, trash and e-waste; where do these come from? Where does the waste from Indian cities go? Let us find out. This article covers how e-waste is generated, its negative impact on the environment and applications of artificial intelligence (AI) in e-waste management.

If you have been around Ghazipur area in New Delhi and looked up, you would have noticed a mountain. Do not think it is Mount Abu; it is a landfill, a mountain of garbage that has accumulated over the years!

New Delhi has many such landfills at various locations including Narela, Bawana, Bhalswa, Okhla and

Ghazipur. The capital city is also home to many other small garbage dumping grounds. Ghazipur landfill is now 65 metres tall, which is only eight metres shorter than Qutub Minar—the tallest minaret of India. The landfill collapsed in September 2017, killing two persons and injuring many others, with cars ending up in the canal. It was a great tragedy. What did we learn from this? Let us delve into it and uncover some facts related to e-waste.

Facts and figures of e-waste

Plastic is present in all electronic products. It is also found in most other things and places. This is where the problem comes in, in the form of pollution. It takes more than 500 years for plastic to decompose.

e-Waste, or electronic/electrical waste, includes discarded computers, pen drives, cellphones, televisions and refrigerators. Discarded electronics contain potentially-harmful materials such as lead, cadmium, beryllium or brominated flame retardants.

Generally, e-waste is categorised into five groups based on materials used: ferrous metals, non-ferrous metals, glass, plastic and others. Nearly 48 per cent of the total e-waste is iron and steel, 21 per cent is plastics, 13 per cent is non-ferrous metals including precious metals of which nearly seven per cent is copper.

Problems of trash and e-waste are not only prevalent in India; these are everywhere including the US. The US and China are the two biggest producers